STAGE 2 CHEMISTRY · MARKING GUIDE

Chemistry Practice Exam Marking Guide

Model answers and mark allocation for the full 120-mark TopMark practice exam

How to use this guide

- Mark your own attempt honestly before checking answers — this is where the real learning happens
- Where a calculation is shown, method marks are available even if the final answer is wrong — check your working against the steps shown
- For extended-response questions, the model answer shows the key points required for full marks; your wording does not need to match exactly
- If you lost marks on a topic, that is exactly the kind of gap the TopMark Crash Course is built to close

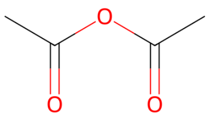
TOPIC 3 · ORGANIC & BIOLOGICAL CHEMISTRY

Question 1 — Aspirin (21 marks)

(a) Carboxylic acid (–COOH) and ester (–O–CO–) functional groups. **2 marks**

1 mark each functional group.

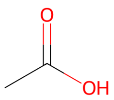
(b)(i) Ethanoic anhydride, $(CH_3CO)_2O$ — two acetyl groups joined by a central oxygen. **1 mark**



(b)(ii) The reaction with anhydride is faster / goes to completion more readily and produces ethanoic acid as a by-product rather than requiring removal of water, giving a higher yield of aspirin. **1 mark**

(c)(i) Hydrolysis (specifically acid-catalysed ester hydrolysis). **1 mark**

(c)(ii) Ethanoic acid, CH_3COOH — drawn showing the –COOH functional group. **2 marks**



1 mark for correct carbon skeleton, 1 mark for correct –COOH group.

(d) Salicylic acid has a free –OH group in addition to its –COOH group, giving it more sites capable of hydrogen bonding with water. Aspirin's –OH has been converted into a less polar ester group, which cannot hydrogen bond with water as effectively. More hydrogen bonding with water means greater solubility, so salicylic acid is more soluble. **3 marks**

1 mark: identifies extra –OH in salicylic acid. 1 mark: links to hydrogen bonding capacity. 1 mark: correct conclusion linking bonding to solubility.

(e) Add a few drops of iron(III) chloride solution. Salicylic acid (a phenol) produces a purple/violet colour; aspirin (no free phenol group) shows no colour change. **2 marks**

1 mark for correct test, 1 mark for correct observation distinguishing the two.

(f) Caffeine would elute first. With a non-polar stationary phase, the less polar compound interacts more strongly with the stationary phase and is retained longer, while the more polar compound has less affinity for the non-polar stationary phase and moves through the column faster. Since caffeine is more polar than aspirin, caffeine has less affinity for the non-polar stationary phase, meaning aspirin (less polar) is retained longer and caffeine elutes first. **3 marks**

Award: 1 mark identifying caffeine elutes first, 1 mark correct polarity reasoning re: stationary phase interaction, 1 mark correct link between weaker interaction and faster elution.

(g) Mass of O in one mole = $4 \times 16.00 = 64.00$ g **3 marks**

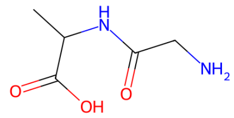
% oxygen = $(64.00 \div 180.16) \times 100 = 35.5\%$

1 mark: correct mass of oxygen (64.00 g). 1 mark: correct method (divide by molar mass, ×100). 1 mark: correct final answer to appropriate sig figs.

Question 1 — Aspirin (21 marks)

continued

(h)(i) Glycine–alanine dipeptide — glycine's carboxyl carbon bonded to alanine's amine nitrogen via a –CO–NH– (amide/peptide) linkage. **2 marks**



1 mark for correct overall dipeptide structure, 1 mark for correctly drawn/labelled peptide bond.

(h)(ii) Two water molecules (one released per peptide bond formed; three amino acids form two linkages). **1 mark**

Total for Question 1: 21 marks

TOPIC 2 · MANAGING CHEMICAL PROCESSES

Question 2 — Haber Process (20 marks)

(a) **5 marks**

ICE table (concentrations, mol L^{-1}, dividing moles by 2.00 L):

N_2: initial 0.60, change −0.20, equilibrium 0.40

H_2: initial 1.80, change −0.60, equilibrium 1.20

NH_3: initial 0, change +0.40, equilibrium 0.40 (0.80 mol ÷ 2.00 L)

$K_c = [NH_3]^2 \div ([N_2][H_2]^3) = (0.40)^2 \div (0.40 \times 1.20^3) = 0.16 \div 0.6912 =$ **0.231 $mol^{-2}\ L^2$** (3 s.f.)

1 mark: correct conversion to concentrations. 1 mark: correct ICE table / equilibrium amounts. 1 mark: correct K_c expression. 1 mark: correct substitution. 1 mark: correct final value. Units are $mol^{-2}\ L^2$, from $M^2 \div (M \times M^3)$.

(b) **3 marks**

Exothermic. As temperature increases, K_c decreases, meaning the equilibrium position shifts toward the reactants (left) as temperature rises. By Le Châtelier's principle, increasing temperature shifts equilibrium in the endothermic direction; since that direction here is the reverse reaction, the forward reaction must be exothermic.

1 mark: correctly reads decreasing K_c trend. 1 mark: correct Le Châtelier reasoning. 1 mark: correct conclusion (exothermic).

(c) **3 marks**

At a lower temperature the reaction rate would be too slow to be commercially viable — even though yield would be higher, it would take far too long to reach equilibrium, making the process economically inefficient. 450°C is a compromise between acceptable yield and acceptable rate.

Accept reasoning based on rate/kinetics compromise. 1 mark: identifies rate issue at low temp. 1 mark: links to economic/commercial viability. 1 mark: frames as compromise between rate and yield.

(d) **2 marks**

Increasing pressure shifts the equilibrium toward the side with fewer moles of gas — in this case, toward the products (2 mol NH_3 vs 4 mol of reactants), increasing the yield of ammonia.

(e) **2 marks**

The catalyst has no effect on the value of K_c (K_c depends only on temperature) but increases the rate at which equilibrium is reached, by providing an alternative reaction pathway with lower activation energy.

(f) **2 marks**

The equilibrium shifts to the right (toward products). Adding N_2 increases the concentration of a reactant, so by Le Châtelier's principle the system shifts to partially consume the added N_2, producing more NH_3 until equilibrium is re-established.

(g) **3 marks**

The yield of ammonia would increase over time. Continuously removing H_2 lowers its concentration below the equilibrium value, so the equilibrium continually shifts right (toward products) to replace some of the removed H_2, producing more NH_3 in the process. Because H_2 is never allowed to reach a stable equilibrium concentration, the reaction is continually driven forward.

1 mark: correct direction (yield increases). 1 mark: correct Le Châtelier reasoning. 1 mark: explains why removal prevents equilibrium being fully re-established, driving further forward reaction.

Total for Question 2: 20 marks

TOPIC 2 · MANAGING CHEMICAL PROCESSES

Question 3 — Contact Process (22 marks)

(a) **3 marks**

The catalyst provides an alternative reaction pathway with a lower activation energy. At a given temperature, a greater proportion of colliding particles have energy exceeding this lower activation energy, so a greater proportion of collisions are successful, increasing the rate of reaction.

Mechanism: the vanadium(V) oxide is reduced and then reoxidised in a cycle, so it is chemically unchanged overall —

$SO_2 + V_2O_5 \rightarrow SO_3 + V_2O_4$, then $2V_2O_4 + O_2 \rightarrow 2V_2O_5$

1 mark: alternative pathway of lower activation energy. 1 mark: links this to a greater proportion of successful collisions. 1 mark: describes the mechanism — either the V_2O_5 reduction/reoxidation cycle above, or adsorption of the gases onto the catalyst surface, reaction there, and desorption of the product.

(b) **1 mark**

At close to atmospheric pressure the conversion of SO_2 to SO_3 is already very high (typically above 98%), so the extra yield gained from operating at high pressure is negligible. High-pressure plant is expensive to build and to run, so the cost is not justified by the very small gain.

Accept either half of the argument: the yield is already near-complete at atmospheric pressure, or the capital/operating cost of high-pressure equipment outweighs the small gain.

(c)(i) **2 marks**

$$SO_2(g) + H_2O(l) \rightarrow H_2SO_3(aq)$$

1 mark: correct reactants/products. 1 mark: correctly balanced with states.

(c)(ii) **2 marks**

$$H_2SO_3(aq) + CaCO_3(s) \rightarrow CaSO_3(s) + H_2O(l) + CO_2(g)$$

Calcium sulfite is insoluble (sulfites are insoluble except those of Group 1 and ammonium), so it is written as (s). Accept the equivalent equation using H_2SO_4/$CaSO_4$(s) if the student assumed full oxidation to sulfuric acid. 1 mark correct products, 1 mark balanced with correct states.

(d) **3 marks**

Increased acidity (lower pH, higher $[H^+]$) shifts the calcium carbonate/shell equilibrium ($CaCO_3 + 2H^+ \rightleftharpoons Ca^{2+} + CO_2 + H_2O$) to the right, favouring dissolution of the shell material. This weakens or dissolves the shells of aquatic organisms, reducing their protection and survival.

1 mark: identifies relevant equilibrium/reaction. 1 mark: correct shift reasoning with increased H^+. 1 mark: links to shell weakening/dissolution.

(e) **2 marks**

340 ppm = 340 mg per kg of gas.

The waste gas density is $1.20\ g\ L^{-1} = 1.20 \times 10^{-3}\ kg\ L^{-1}$.

Concentration = $340\ mg\ kg^{-1} \times 1.20 \times 10^{-3}\ kg\ L^{-1}$ = **$0.408\ mg\ L^{-1}$**

1 mark: correct method (ppm × density). 1 mark: correct final answer. Note the density is given per litre, not per millilitre — a waste gas is roughly a thousand times less dense than a liquid.

(f) **2 marks**

Capturing and reacting the SO_2 converts it into saleable sulfuric acid rather than losing it as waste, increasing the manufacturer's product yield and revenue from the same input of raw material. It may also avoid fines/costs associated with emissions regulations.

Question 3 — Contact Process (22 marks)

continued

(g) Advantage: double absorption increases the overall conversion of SO_2 to SO_3 (typically from ~98% to over 99.5%), reducing SO_2 emissions and associated environmental harm (acid rain) — linking to the SHE concept of *Development of science understanding relies on and advances technology, and technology relies on and advances science*, or *Applications and limitations*. Limitation: the additional absorption step requires extra equipment and energy, increasing the capital and operating cost of the plant, which may not be economically justified for smaller producers — linking to *Applications and limitations of science*. **4 marks**

1 mark identifying a real advantage, 1 mark linking it to a SHE concept, 1 mark identifying a real limitation, 1 mark linking it to a SHE concept.

(h) Fertiliser manufacture (e.g. production of ammonium sulfate or superphosphate). Other acceptable answers: car battery electrolyte, detergent manufacture, ore processing. **1 mark**

(i) **2 marks**

$$N_2(g) + O_2(g) \rightarrow 2NO(g)$$

1 mark correct formulas, 1 mark correctly balanced.

Total for Question 3: 22 marks · End of Booklet 1: 63 marks

TOPIC 1 · MONITORING THE ENVIRONMENT

Question 4 — AAS & Titration (15 marks)

(a) Reading from the calibration graph at absorbance 0.62: approximately **0.39 mg L^{-1}** (accept 0.37–0.40, or any value consistent with the drawn line of best fit). **1 mark**

(b) Electrons in lead atoms exist at specific, quantised energy levels. Only photons with energy exactly matching the gap between two energy levels can be absorbed (promoting an electron to an excited state); since these energy gaps are unique to lead's electron configuration, only specific wavelengths are absorbed. **2 marks**

(c) Using ~0.39 mg L^{-1} from (a): 0.39 mg L^{-1} × 1000 = **390 µg L^{-1}**. This is far above the 10 µg L^{-1} guideline (about 39 times it), so yes, this sample exceeds the guideline. **2 marks**

1 mark: correct unit conversion (×1000). 1 mark: correct comparison/conclusion, consistent with the student's part (a) value.

(d) Accuracy. **1 mark**

(e) Standard addition involves adding known, increasing amounts of the analyte (lead) directly to portions of the sample itself, rather than using separately prepared standards. Because each spiked portion contains the same sample matrix as the original, any interference from other dissolved ions affects the standards and the sample equally, cancelling out the matrix effect. Plotting absorbance against added concentration and extrapolating back to zero absorbance gives the original sample's concentration corrected for interference. **3 marks**

1 mark: describes adding known amounts to the sample itself. 1 mark: explains why this cancels matrix interference. 1 mark: correct method for determining final concentration (extrapolation to x-intercept).

(f)(i) **1 mark**

$$Ag^{+}(aq) + Cl^{-}(aq) \rightarrow AgCl(s)$$

(f)(ii) **3 marks**

$n(Ag^{+}) = 0.0500 \text{ mol L}^{-1} \times 0.01420 \text{ L} = 7.10 \times 10^{-4}$ mol

Since Ag^{+} : Cl^{-} is 1:1, $n(Cl^{-}) = 7.10 \times 10^{-4}$ mol

$[Cl^{-}] = 7.10 \times 10^{-4} \text{ mol} \div 0.02500 \text{ L} = 0.0284 \text{ mol L}^{-1}$

1 mark: correct moles $AgNO_3$. 1 mark: correct 1:1 mole ratio applied. 1 mark: correct final concentration.

(f)(iii) **2 marks**

Mass concentration = 0.0284 mol L^{-1} × 35.45 g mol^{-1} = 1.0068 g L^{-1}

= **1.01 g L^{-1} = 1010 mg L^{-1}** (3 s.f., matching the 3 s.f. concentration from part (ii))

1 mark: correct method (× molar mass of Cl, convert to mg). 1 mark: correct final value given to an appropriate number of significant figures.

Total for Question 4: 15 marks

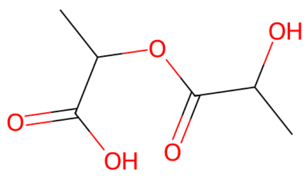
TOPIC 3 · ORGANIC & BIOLOGICAL CHEMISTRY

Question 5 — PLA (14 marks)

(a) Hydroxyl (–OH) and carboxylic acid (–COOH) functional groups. **2 marks**

(b) Condensation polymerisation; each ester linkage formed releases one water molecule. **2 marks**

(c) PLA dimer, two lactic acid units joined by an ester linkage (–CO–O–). **2 marks**



1 mark correct monomer units shown, 1 mark correct ester linkage between them.

(d) PLA contains ester linkages, which can be hydrolysed by water/enzymes under composting conditions, breaking the polymer back into lactic acid monomers. Polyethylene's backbone is a long chain of C–C bonds with no hydrolysable functional groups, so it cannot be broken down by water or microorganisms and persists in the environment. **3 marks**

(e) Mass contributed by 1200 monomer units = 1200 × 90.08 = 108 096 g mol^{-1} **3 marks**

1200 monomers form 1199 ester linkages, so 1199 water molecules are released: 1199 × 18.02 = 21 606 g mol^{-1}

Average molar mass = 108 096 − 21 606 = 86 490 ≈ **8.65 × 10^4 g mol^{-1}**

Accept the approximation using n rather than n−1 linkages for a chain this long (1200 × 18.02 = 21 624, giving 86 472 — also ≈ 8.65 × 10^4). 1 mark: correct mass of monomer units. 1 mark: correct mass of water removed. 1 mark: correct final subtraction and answer.

(f) With reference to *Applications and limitations of science*: while PLA is biodegradable in industrial composting conditions, it typically does not break down effectively in a standard landfill or the natural environment (which lack the heat/moisture of industrial composting), meaning the environmental benefit depends on proper waste infrastructure being available — a limitation not always communicated to consumers. **2 marks**

Total for Question 5: 14 marks

TOPIC 4 · MANAGING RESOURCES

Question 6 — Copper Refining (14 marks)

(a)(i)	$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$	**1 mark**
(a)(ii)	Both the anode and the product deposited at the cathode are the same element (copper) — impure copper dissolves at the anode and pure copper is deposited at the cathode, so the process purifies existing copper rather than producing copper from a different starting material (as in electrolytic production from an ore/molten compound).	**2 marks**
(a)(iii)	Cu: [Ar] $3d^{10}$ $4s^1$; removing 2 electrons ($4s^1$ and one 3d electron) gives: Cu^{2+}: $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $3d^9$ *1 mark: correct core configuration up to $3p^6$. 1 mark: correct $3d^9$ (not 4s-retained).*	**2 marks**
(b)(i)	$n(S_2O_3^{2-})$ = 0.1000 mol L^{-1} × 0.01840 L = 1.840 × 10^{-3} mol	**1 mark**
(b)(ii)	From the second equation, I_2 : $S_2O_3^{2-}$ is 1:2, so $n(I_2)$ = 1.840 × 10^{-3} ÷ 2 = 9.20 × 10^{-4} mol. From the first equation, Cu^{2+} : I_2 is 2:1, so $n(Cu^{2+})$ = 9.20 × 10^{-4} × 2 = 1.840 × 10^{-3} mol. $[Cu^{2+}]$ = 1.840 × 10^{-3} mol ÷ 0.02000 L = **0.0920 mol L^{-1}** (3 s.f.) *1 mark: correct I_2 moles via 1:2 ratio. 1 mark: correct Cu^{2+} moles via 2:1 ratio. 1 mark: correct division by volume. 1 mark: correct final answer to appropriate sig figs.*	**4 marks**
(c)	Silver and gold are less reactive than copper (lower in the activity series), so they are not oxidised at the anode under conditions that oxidise copper; they simply fall away as solid anode sludge. Because silver and gold are valuable precious metals, recovering and selling this sludge provides a significant additional revenue stream for the refinery, sometimes covering much of the cost of the refining process.	**3 marks**
(d)	Fuel cells / hydrogenation of oils to make margarine / ammonia production (Haber process) / hydrogen peroxide manufacture, or other valid commercial use. *The chlor-alkali process electrolyses aqueous sodium chloride, giving H_2 at the cathode, Cl_2 at the anode and NaOH in solution. Note that hydrogen is not produced in the electrolytic production of a reactive metal such as aluminium — the Hall–Héroult cell uses molten Al_2O_3 in cryolite, with no water present.*	**1 mark**

Total for Question 6: 14 marks

TOPIC 1 · MONITORING THE ENVIRONMENT

Question 7 — Eutrophication (14 marks)

(a) Excess nitrate acts as a nutrient that causes rapid, excessive growth of algae (an algal bloom) on the water surface. This blocks sunlight from reaching plants below, and when the algae die, decomposing bacteria consume large amounts of dissolved oxygen as they break down the dead algae. The resulting drop in dissolved oxygen (hypoxia) suffocates fish and other aquatic organisms. **3 marks**

(b) Positive charge (an anion-exchange resin is needed to attract and remove the negatively charged nitrate ion). **1 mark**

(c) 62 ppm = 62 mg L^{-1} (since density ≈ 1.00 g mL^{-1}). **2 marks**
$M(NO_3^-) = 14.01 + (3 \times 16.00) = 62.01$ g mol^{-1}
n = mass ÷ molar mass = 0.062 g ÷ 62.01 g mol^{-1} = **1.0×10^{-3} mol L^{-1}**

1 mark: correct conversion to mg/L then g/L. 1 mark: correct final concentration using the NO_3^- molar mass (62.01 g mol^{-1}).

(d) Common calcium and aluminium nitrate salts are highly soluble in water (most nitrate salts are soluble), so adding Ca^{2+} or Al^{3+} does not cause nitrate to precipitate out of solution, unlike phosphate, which forms insoluble calcium/aluminium phosphate precipitates. **2 marks**

(e) Balancing atoms, charge, and electrons for the reduction of N (+5 in NO_3^-) to N (0 in N_2): **3 marks**

$$2NO_3^- + 12H^+ + 10e^- \rightarrow N_2 + 6H_2O$$

1 mark: correct N and O balance ($2NO_3^-$ to N_2, plus H_2O). 1 mark: correct H balance using H^+. 1 mark: correct electron balance ($10e^-$, consistent with 2 N atoms each gaining 5 electrons).

(f) With reference to *Science and society* or *Applications and limitations*: stricter fertiliser regulation could reduce crop yields and increase costs for farmers in the short term as they adjust to lower fertiliser inputs or invest in more precise application technology, potentially affecting farm profitability and food prices, even though the regulation provides a long-term environmental benefit to waterways and the communities that rely on them. **3 marks**

Total for Question 7: 14 marks · End of Booklet 2: 57 marks

HOW DID YOU GO?

Knowing where you lost marks is the whole point.

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